

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0897

Kearneyville 345 kV
January 22, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0897. The Interconnection Customer proposes a 463.4 MW natural gas combined cycle facility interconnecting at the Kearneyville 345 kV substation in the West Virginia area. The request was received on November 23, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 242 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$71,640,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ros Fork – G-P Gypsum Substation 138.0 kV
Contingency: Air Force Base – Artesia Country Club 345.0 kV
Loading: 151.9% (Rating: 259 MVA)
DFAX: 8.0%
Pre-project loading: 137.6%
2. Facility: Bassett – Monocacy 138.0 kV
Contingency: Artesia Country Club – Clark County Substation 345.0 kV
Loading: 151.0% (Rating: 217 MVA)
DFAX: 5.9%
Pre-project loading: 138.4%
3. Facility: Monocacy – Artesia Country Club 138.0 kV
Contingency: Artesia Country Club – Clark County 345.0 kV
Loading: 148.1% (Rating: 215 MVA)
DFAX: 5.9%
Pre-project loading: 135.5%
4. Facility: Walkersville – Bassett 138.0 kV
Contingency: Clark County – Artesia Country Club 345.0 kV Line
Loading: 137.7% (Rating: 258 MVA)
DFAX: 5.9%
Pre-project loading: 127.2%
5. Facility: Edgewood – Walkersville 138.0 kV
Contingency: Artesia Country Club Substation – Clark County 345.0 kV Line
Loading: 123.1% (Rating: 306 MVA)
DFAX: 7.2%
Pre-project loading: 112.2%

6. Facility: Rochester Hydro – Deere 138.0 kV
Contingency: Kearneyville – Clark County 345.0 kV
Loading: 120.4% (Rating: 212 MVA)
DFAX: 8.0%
Pre-project loading: 103.0%
7. Facility: Clover Vly – Ross Fork 138.0 kV
Contingency: Air Force Base – Artesia Country Club 345.0 kV Line
Loading: 114.8% (Rating: 301 MVA)
DFAX: 6.6%
Pre-project loading: 104.7%
8. Facility: Artesia Country Club 345-1 kV Autotransformer
Contingency: Artesia Country Club – Air Force Base 345.0 kV Line
Loading: 109.6% (Rating: 540 MVA)
DFAX: 18.9%
Pre-project loading: 93.3%
9. Facility: Cholame – Rochester Hydro 138.0 kV L
Contingency: Kearneyville – Clark County 345.0 kV
Loading: 109.1% (Rating: 239 MVA)
DFAX: 6.1%
Pre-project loading: 97.3%
10. Facility: Edgewood – Five Channels Substation 138.0 kV
Contingency: Artesia Country Club – Clark County 345.0 kV line
Loading: 109.0% (Rating: 369 MVA)
DFAX: 7.2%
Pre-project loading: 100.0%
11. Facility: Artesia Country Club – Ross Fork 138.0 kV line
Contingency: Air Force Base – Artesia Country Club 345.0 kV
Loading: 104.3% (Rating: 647 MVA)
DFAX: 18.3%
Pre-project loading: 91.3%
12. Facility: Riverton – Front Royal 138.0 kV
Contingency: Clark County – Kearneyville Substation 345.0 kV
Loading: 103.8% (Rating: 250 MVA)
DFAX: 5.8%
Pre-project loading: 93.1%
13. Facility: Artesia Country Club 138 kV/1 kV TX
Contingency: Air Force Base – Artesia Country Club 345.0 kV
Loading: 102.9% (Rating: 575 MVA)
DFAX: 18.9%
Pre-project loading: 87.6%
14. Facility: Smallwood – Columbus City 138.0 kV L
Contingency: Clark County – Kearneyville 345.0 kV Line
Loading: 100.5% (Rating: 231 MVA)
DFAX: 11.0%
Pre-project loading: 78.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ros Fork – G-P Gypsum Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$7,961,000	100% (\$7,961,000)
Bassett – Monocacy 138.0 kV	Rebuild line with higher-capacity conductor	\$3,887,000	100% (\$3,887,000)
Monocacy – Artesia Country Club 138.0 kV	Rebuild line with higher-capacity conductor	\$11,381,000	100% (\$11,381,000)
Walkersville – Bassett 138.0 kV	Rebuild line with higher-capacity conductor	\$1,753,000	100% (\$1,753,000)
Edgewood – Walkersville 138.0 kV	Rebuild line with higher-capacity conductor	\$3,949,000	100% (\$3,949,000)
Depere – Rochester Hydro 138.0 kV	Rebuild line with higher-capacity conductor	\$27,574,000	46% (\$12,763,000)
Clover Vly – Ross Fork 138.0 kV	Rebuild line with higher-capacity conductor	\$3,796,000	100% (\$3,796,000)
Artesia Country Club 345-1 kV Autotransformer	Replace transformer with higher-rated unit	\$16,242,000	84% (\$13,652,000)
Rochester Hydro – Cholame 138.0 kV	Reconductor line	\$2,990,000	100% (\$2,990,000)
Edgewood – Five Channels 138.0 kV	Reconductor line	\$1,075,000	100% (\$1,075,000)
Artesia Country Club – Ross Fork 138.0 kV line	Reconductor line	\$1,462,000	77% (\$1,121,000)
Riverton – Front Royal 138.0 kV	Reconductor line	\$1,232,000	100% (\$1,232,000)
Artesia Country Club 138/1kV XFMR	Replace transformer with higher-rated unit	\$5,284,000	84% (\$4,441,000)
Columbus City – Smallwood 138.0 kV	Reconductor line	\$1,639,000	100% (\$1,639,000)
	Total allocated to project		\$71,640,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.